

Unit: 3 Matrices

Page No.

Date

Defⁿ, types of matrices: unit matrix, UTM, LTM, Diagonal matrix, Row matrix, column matrix, sq. matrix, Multiplication of matrices, transpose of matrix,

Rank of Matrix: $\Rightarrow$ A matrix is said to be of rank $\geq$ when i) it has at least one non-zero minor of order $\geq$ & ii) every minor of order higher than $\geq$ vanishes. (OR) NO. of non-zero rows in UTM or LTM determine the rank of matrix.

ex. ① $A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 4 & 2 \\ 2 & 6 & 5 \end{bmatrix}$ All 3rd order minor of $A = 0$
but 2nd order minor of $A \neq 0$
i.e. $\begin{vmatrix} 1 & 2 \\ 1 & 4 \end{vmatrix} = 2 \neq 0$

$\therefore S(A) = 2$

② $A = \begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix}$

$R_1 \leftrightarrow R_2$

$A \sim \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -3 & -1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix}$ $R_3 \rightarrow R_3 - 3R_1$
 $R_4 \rightarrow R_4 - R_1$

$A \sim \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -3 & -1 \\ 0 & 1 & -3 & -1 \\ 0 & 1 & -3 & -1 \end{bmatrix}$ $R_3 \rightarrow R_3 - R_2$ $R_4 \rightarrow R_4 - R_2$ $\sim \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -3 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

$R_3 \rightarrow R_3 + 2R_2$ No. of non-zero rows = 2
 $\therefore S(A) = 2$

Echelon Form

The row echelon form of the matrix is

$$\begin{bmatrix} z_{11} & z_{12} & \dots & z_{1n} \\ 0 & z_{22} & \dots & z_{2n} \\ 0 & 0 & z_{33} & \dots & z_{3n} \\ 0 & 0 & 0 & \dots & 0 \\ 0 & 0 & 0 & \dots & 0 \end{bmatrix}$$

In it, rows of zeros, if present, are the last rows & in each non-zero row the uppermost non-zero entry is farther to the right than in the previous row.

e.g. $\begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & 4 \\ 0 & 0 & 0 \end{bmatrix}$

Soln of system of eqns & consistency of linear system of eqns :-

Procedure to test the consistency & soln of linear system of eqns :-

let $\begin{aligned} a_{11}x + a_{12}y + a_{13}z &= b_1 \\ a_{21}x + a_{22}y + a_{23}z &= b_2 \\ a_{31}x + a_{32}y + a_{33}z &= b_3 \end{aligned}$

step ① write matrix form i.e. $Ax = B$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

$\downarrow \qquad \qquad \downarrow \qquad \qquad \downarrow$
 $A \qquad \qquad X \qquad = \qquad B$
 coefficient matrix column matrix of unknown column matrix of constant

step ②: write Augmented matrix as $[A:B]$

$$[A:B] = \begin{bmatrix} a_{11} & a_{12} & a_{13} & : & b_1 \\ a_{21} & a_{22} & a_{23} & : & b_2 \\ a_{31} & a_{32} & a_{33} & : & b_3 \end{bmatrix}$$

Page No.

Date

step ③: using Row elementary transformⁿ convert Augmented matrix into Row echelon form or upper triangular matrix.

$$[A:B] \sim \begin{bmatrix} a_{11} & a_{12} & a_{13} & : & d_1 \\ 0 & a'_{22} & a'_{23} & : & d_2 \\ 0 & 0 & a'_{33} & : & d_3 \end{bmatrix}$$

$\downarrow$ $\downarrow$
A B

step ④: $\Rightarrow$

case ① If $\text{rank of } A = \text{rank of } [A:B] = n$
[i.e. $a'_{33} \neq 0$ ~~known condition~~]
then the system is consistent & no. of unknown
that linear system have Unique solⁿ

case ② If $a'_{33} = 0$ & $d_3 = 0$ then
 $\text{Rank of } A = \text{Rank of } [A:B] < n$
system is consistent & have infinite solⁿ

case ③ if $a'_{33} = 0$ & $d_3 \neq 0$ then
 $\text{Rank of } A \neq \text{Rank of } [A:B]$

$\therefore$ system is inconsistent & have no solⁿ

① Test the consistency & solve

$$5x + 3y + 7z = 4, \quad 3x + 26y + 2z = 9, \quad 7x + 2y + 10z = 5$$

soln, we have $Ax = B$

$$\begin{bmatrix} 5 & 3 & 7 \\ 3 & 26 & 2 \\ 7 & 2 & 10 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 9 \\ 5 \end{bmatrix}$$

The Augmented matrix is

$$[A:B] = \begin{bmatrix} 5 & 3 & 7 & : & 4 \\ 3 & 26 & 2 & : & 9 \\ 7 & 2 & 10 & : & 5 \end{bmatrix}$$

$$R_2 \rightarrow 5R_2 - 3R_1, \quad R_3 \rightarrow 5R_3 - 7R_1$$

$$[A:B] \sim \begin{bmatrix} 5 & 3 & 7 & : & 4 \\ 0 & 121 & -11 & : & 33 \\ 0 & -11 & 1 & : & -3 \end{bmatrix}$$

$$[A:B] \sim \begin{bmatrix} 5 & 3 & 7 & : & 4 \\ 0 & 11 & -1 & : & 3 \\ 0 & -11 & 1 & : & -3 \end{bmatrix} \quad \begin{array}{l} R_2 \rightarrow \frac{1}{11} R_2 \\ R_3 \rightarrow R_3 + R_2 \end{array}$$

$$R_3 \rightarrow R_3 + R_2$$

$$[A:B] \sim \begin{bmatrix} 5 & 3 & 7 & : & 4 \\ 0 & 11 & -1 & : & 3 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$$

$$\text{Here } \rho(A) = \rho(A:B) = 2 = r < n (=3)$$

$\therefore$ system is consistent but system have infinite soln as $r < n$

Now $Ax = B$

$$\begin{bmatrix} 5 & 3 & 7 \\ 0 & 11 & -1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \\ 0 \end{bmatrix}$$

let ~~z=k~~

$$\therefore 5x + 3y + 7z = 4 \quad \text{--- (1)}$$

$$11y - z = 3 \quad \text{--- (2)}$$

let $y = k$

$$\therefore 11k - z = 3$$

$$z = 11k - 3 \quad \text{use in eqn (1)}$$

$$\text{(1)} \Rightarrow 5x + 3k + 7(11k - 3) = 4$$

$$5x + 3k + 77k - 21 = 4$$

$$5x + 80k = 25$$

$$5x = 25 - 80k$$

$$x = \frac{5(5 - 16k)}{5}$$

$$x = 5 - 16k$$

(2) Investigate the value of d & u so that the eqns $2x + 3y + 5z = 9$, $7x + 3y - 2z = 8$, $2x + 3y + dz = u$ have (i) no soln (ii) a unique soln (iii) an infinite no. of soln

$\Rightarrow$ we have $Ax = B$

$$\begin{bmatrix} 2 & 3 & 5 \\ 7 & 3 & -2 \\ 2 & 3 & d \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 9 \\ 8 \\ u \end{bmatrix}$$

The Augmented matrix is $[A:B]$

$$[A:B] = \begin{bmatrix} 2 & 3 & 5 & : & 9 \\ 7 & 3 & -2 & : & 8 \\ 2 & 3 & d & : & u \end{bmatrix}$$

$$R_2 \rightarrow 2R_2 - 7R_1; \quad R_3 \rightarrow R_3 - R_1$$

$$[A:B] = \begin{bmatrix} 2 & 3 & 5 & : & 9 \\ 0 & -15 & -39 & : & -47 \\ 0 & 0 & d-5 & : & u-9 \end{bmatrix}$$

~~if $x=5$ & $y=9$~~

(i) if $d-5=0$ i.e. $d=5$ & $u-9 \neq 0$ i.e. $u \neq 9$
then $S(A) \neq S(A:B)$

$\therefore$ system have no solⁿ

(ii) if $d-5 \neq 0$ i.e. $d \neq 5$ & $u-9 = \text{any}$ then

$$S(A) = S(A:B) = 3 = n$$

$\therefore$ system have unique solⁿ

(iii) if $d-5=0$ i.e. $d=5$ & $u-9=0$ i.e. $u=9$
then $S(A) = S(A:B) = 2 < n$

$\therefore$ system have infinite solⁿ no. of solⁿ

Practice Problems: $\Rightarrow$

1) Test the consistency & solve them if consistent

(1) $x-2y+3z=3$, $2x+y+z+t=-4$, $4x-3y+z+7t=8$

(2) $4x-2y+6z=8$, $x+y-3z=-1$, $15x-3y+9z=21$

(3) $2x-3y+7z=5$, $3x+y-3z=13$, $2x+19y-47z=32$

(4) $x+2y+z=3$, $2x+3y+2z=5$, $3x-5y+5z=2$,
 $3x+9y-z=4$

(5) $2x+6y+11=0$, $6x+20y-6z+3=0$, $6y-18z+1=0$

(6) $3x+3y+2z=1$, $x+2y=4$, $10y+3z=-2$.
 $2x-3y-z=5$

2) Investigate the value of d & u for which

$$x+y+z=6, x+2y+3z=10, x+2y+dz=u$$

have (i) no solⁿ (ii) a unique solⁿ (iii) an infinite no. of solⁿ

3) Find the value of a & b for which the eq^s

$$x+ay+z=3, x+2y+2z=b, x+5y+3z=9$$

are consistent. when will these eq^s have a unique solⁿ.

4) S.T. the if $d \neq 5$, the system of eqn
 $3x - y + 4z = 3$, $x + 2y - 3z = -2$, $6x + 5y + dz = -3$
 have unique soln. If $d = -5$, show that the eqns
 are consistent. Determine the soln in each case.

5) S.T. the eqn $3x + 4y + 5z = a$, $4x + 5y + 6z = b$
 $5x + 6y + 7z = c$ do not have a soln unless $a + c = 2b$
 i.e. a, b, c are in A.P.

soln $\Rightarrow$

$$[A:B] = \begin{bmatrix} 3 & 4 & 5 & : & a \\ 4 & 5 & 6 & : & b \\ 5 & 6 & 7 & : & c \end{bmatrix}$$

$$b - a = c - b$$

$$\frac{a}{b} + \frac{c}{b} = 2$$

$$2b = a + c$$

a, b, c in A.P.

$$\frac{b}{a} = \frac{c}{b} = k$$

$$R_2 \rightarrow 3R_2 - 4R_1, \quad R_3 \rightarrow 3R_3 - 5R_1$$

$$[A:B] = \begin{bmatrix} 3 & 4 & 5 & : & a \\ 0 & -1 & -2 & : & 3b - 4a \\ 0 & -2 & -4 & : & 3c - 5a \end{bmatrix}$$

$$R_3 \rightarrow -1R_3 \quad R_2 \rightarrow -R_2$$

$$[A:B] = \begin{bmatrix} 3 & 4 & 5 & : & a \\ 0 & 1 & 2 & : & 4a - 3b \\ 0 & 2 & 4 & : & 5a - 3c \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 2R_2 \quad R_3 \rightarrow R_3 - 2$$

$$5a - 3c - 8a + 6b$$

$$[A:B] = \begin{bmatrix} 3 & 4 & 5 & : & a \\ 0 & 1 & 2 & : & 4a - 3b \\ 0 & 0 & 0 & : & -3a + 6b - 3c \end{bmatrix}$$

$$R_3 \rightarrow \frac{1}{3}R_3$$

$$[A:B] = \begin{bmatrix} 3 & 4 & 5 & : & a \\ 0 & 1 & 2 & : & 4a - 3b \\ 0 & 0 & 0 & : & -a + 2b - c \end{bmatrix}$$

if $-a + 2b - c = 0$ i.e. $2b = a + c$ then $S(A) = S(A:B) = 2$
 $\therefore$ system have infinite no. of soln.